SYNOPSIS

Autonomous Video Ad-Placement & Contextual Safety Analyzer

Submitted in partial fulfillment of the requirements for the course

Specialization Project

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1. Abstract

Digital content creation and marketing agencies face significant revenue losses due to the demonetization of videos containing accidental unsafe content or unapproved branding. Furthermore, identifying optimal spatial and temporal zones for digital ad insertions requires tedious manual review. The proposed Autonomous Video Ad-Placement & Contextual Safety Analyzer is an offline, desktop-based multimodal AI system designed to solve these bottlenecks. By leveraging Natural Language Processing (NLP) for audio transcription and Computer Vision (CV) for spatial frame analysis, the system automatically flags compliance risks and identifies premium ad-placement surfaces in real-time. Operating entirely on local hardware, the solution ensures strict data privacy for unreleased media while significantly reducing post-production auditing time.

2. Problem Statement

Content creators and media agencies spend excessive hours manually reviewing long-form video files to ensure compliance with strict community guidelines and to locate "blank spaces" suitable for digital sponsor overlays. Utilizing cloud-based AI APIs for this process is prohibitively expensive for small businesses and risks leaking unreleased, proprietary media. There is a critical need for a locally hosted, automated auditing pipeline that can simultaneously analyze video and audio tracks to optimize monetization workflows without compromising data privacy.

3. Objectives

1. To automate the detection of contextual safety violations (profanity, sensitive topics) using local NLP.
2. To identify and log premium spatial zones (e.g., blank walls, empty screens) for ad insertions using object detection.
3. To eliminate the reliance on costly, privacy-compromising cloud APIs by executing all models locally.
4. To consolidate multimodal metadata into a centralized relational database.
5. To generate an interactive, timeline-based dashboard for rapid video review and metadata export.

4. Existing System

Currently, most digital marketing agencies and independent creators rely on:

- Manual frame-by-frame review by video editors.
- Basic keyword-filtering tools that lack semantic understanding.
- Expensive enterprise cloud platforms (e.g., AWS Rekognition) that require massive bandwidth to upload raw 4K footage.

These traditional approaches are highly inefficient, subjective, and operationally expensive for localized production teams.

5. Limitations of Existing Systems

- Manual compliance auditing is severely time-consuming and prone to human oversight.
- Cloud-based processing incurs heavy recurring subscription fees and high bandwidth costs.
- Uploading unreleased client footage to external servers often violates strict Non-Disclosure Agreements (NDAs).
- Existing tools rarely synchronize spatial CV data with audio NLP data in a unified timeline.

6. Proposed Solution

The proposed solution is an Edge-AI desktop application that processes multimedia files sequentially to optimize system memory. The system will:

- Extract and transcribe audio tracks locally using OpenAI Whisper to flag semantic compliance risks.
- Sample video frames and utilize YOLOv8 to track spatial environments and detect visual anomalies.
- Cross-reference detected safe zones to log continuous 5-second "Premium Ad Spots."
- Store all extracted timestamps and coordinates in a local SQLite database.
- Render the findings on an interactive HTML/Streamlit dashboard for the end-user.

7. System Functionalities

1. Audio Extraction and NLP Sentiment Module
2. Video Frame Sampling and CV Object Detection Engine
3. Compliance Ruleset Matching Algorithm
4. SQLite Database Integration and Logging
5. Streamlit Interactive UI Dashboard
6. Automated Metadata Export Module

8. Tools and Technologies

Technology	**Purpose**
Python	Core Pipeline Development
OpenCV & MoviePy	Video Frame and Audio Extraction
YOLOv8	Computer Vision / Object Detection
OpenAI Whisper	Local Audio-to-Text Transcription
SQLite	Local Relational Database
Streamlit	Interactive Frontend Dashboard
GitHub	Version Control and Collaboration

9. Expected Outcomes

- Drastic reduction in the time required for pre-publishing video audits.
- Increased monetization potential through the automated discovery of ad-placement inventory.
- Complete data sovereignty and privacy, as no media files are transmitted over the internet.
- A scalable, low-cost alternative to enterprise media processing suites.

10. Project Timeline

Week	Activity
1–2	Requirement Analysis and Environment Setup
3–4	Development of MoviePy Audio Pipeline and Whisper Integration
5–7	OpenCV Frame Sampling and YOLOv8 Customization
8–9	Database Schema Design and Integration
10–11	Streamlit Dashboard Development and System Testing
12	Final Debugging, Documentation, and Review

11. Conclusion

The Autonomous Video Ad-Placement & Contextual Safety Analyzer represents a significant step forward in edge-computing applications for digital media. By merging Computer Vision and Natural Language Processing into a unified, locally hosted pipeline, the project provides a secure, highly efficient, and commercially viable solution to modern video compliance and monetization challenges.

12. References

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